

STACK

Q10)

Question:

DS	Session	Stack	Question Information	Level 1	Challenge 50
Question description Hassan gets a job in a software company in Hyderabad. The training period for the first three months is 20000 salary. Then incremented to 25000 salaries. Training is great but they will give you a programming task every day in three months. Hassan must finish it in the allotted time. His teammate Jocelyn gives him a task to complete the concept of Postfix to Prefix Conversion for a given expression. can you help him? Functional Description: • Read the Prefix expression in reverse order (from right to left) • If the symbol is an operand, then push it onto the Stack • If the symbol is an operator, then pop two operands from the Stack Create a string by concatenating the two operands and the operator after them. string = operand1 + operand2 + operator And push the resultant string back to Stack • Repeat the above steps until end of Prefix expression. Constraints the input should be a expressions					

PROGRAM:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char exp[100], stack[100][100];
6      int top = -1;
7
8      scanf("%s", exp);
9
10     for (int i = 0; exp[i] != '\0'; i++) {
11         char c = exp[i];
12
13         if ((c >= 'A' && c <= 'Z') || (c >= 'a' && c <= 'z')) {
14             stack[++top][0] = c;
15             stack[top][1] = '\0';
16         } else {
17             char a[100], b[100], temp[100];
18             strcpy(a, stack[top--]);
19             strcpy(b, stack[top--]);
20
21             sprintf(temp, "%c%s%s", c, b, a);
22             strcpy(stack[++top], temp);
23         }
24     }
25
26     printf("%s", stack[top]);
27
28     return 0;
29 }
```

OUTPUT:

TEST CASE 1:

```
ABC/-AK/L-*  
*-A/BC-/AKL
```

TEST CASE 2:

```
AB+CD-*  
*+AB-CD
```